

Lecture 6 (9/25)

Variance, Normals models and Linear Regression

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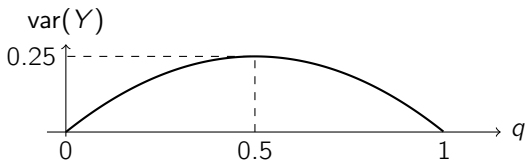
In code `np.var(y)`. The standard deviation is $\sigma_Y = \sqrt{\text{var}(Y)}$

Example

Let $Y \sim \text{Bernoulli}(q)$. Compute $\text{var}(Y)$.

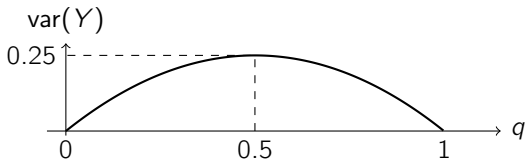
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Let $Y \sim \text{Uniform}(0, L)$. Compute $\text{var}(Y)$.

The Normal distribution

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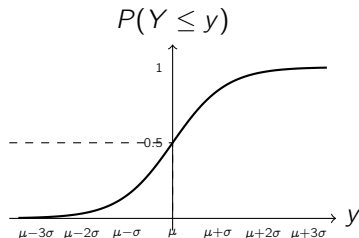
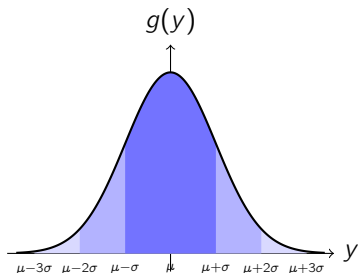
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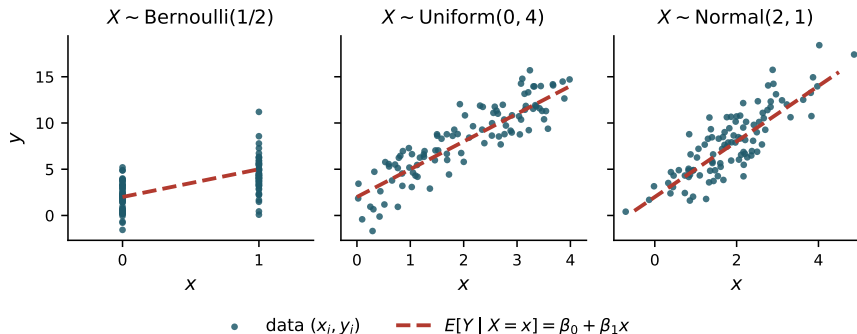
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- X is the predictor
- Y is the response variable
- β_1 is called the regression coefficient of X .

What data from this model looks like

Data simulated from $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma^2)$



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- Exogeneity: ϵ is independent of X .

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This formula connects the regression coefficient the covariance (and hence expectation we can compute from data).

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Consider the standardized variables

$$Z_X = (X - \mu_X)/\sigma_X, \quad Z_Y = (Y - \mu_Y)/\sigma_Y$$

What is the regression coefficient of Z_X on Z_Y ?